

Quantum Simulation of Allosteric Signal Propagation

residue-level connectivity maps and hit lists from a derived force network, measured at 156 qubits

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-cleveland-gic2026>

Summary

0.71–0.89

residue-level AUC against the holo-defined allosteric site on cardiac myosin, BCR-ABL1, CDK2 and glucokinase (2,000-trial permutation $p = 0.0005$), above the strongest structure-only floor on CDK2 and glucokinase, with the controller/receiver split landing where the biology says it should; top-5 hit lists per Table-1 target and a frozen blind prediction for c-Myc

8.7σ

long-range connected correlation $C(28-11) = -0.0757 \pm 0.0087$ measured at lattice distance 11 on a 156-qubit $K=6$ map, pre-registered, $K=2$ anchor exact — a propagation quantity whose exact classical counterpart spans a 99-qubit connected cone

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targets with the mediated-response field solved natively on a photonic machine at cosine 0.965–0.997 to the exact field; cross-paradigm anchors on neutral-atom and a second superconducting vendor

Why the numbers can be trusted

The force network is derived rather than fitted: Coulomb at derived α , a derived 0.21 eV hydrogen-bond quantum, site energies at the kT rung; the same layer predicts SKEMPI double-mutant couplings with zero free parameters. Every hardware map is pre-registered (rule and predictions hashed before data), anchored at $K=2$ against exact light-cone computation, and carries in-map null controls. Metrics were frozen before the fresh-target validation; the null-model correction is disclosed as method.

This is a Phase-1 concept proposal in the seven guideline sections. Prior work is receipted (job IDs and archived counts, negatives kept); the Phase-2 PoC delivers $K=6$ maps for every Table-1 target with the residue-level AUC as the pre-registered endpoint.

Concept proposal follows (§1–§7, appendices A–B) ·

<https://github.com/sharadbachani-oss/merlin-quantum-cleveland-gic2026>

Concept Proposal

1. Problem framing

Allosteric regulation is signal propagation through a protein's contact network at the thermal rung where the coupling energy is of order 100 kT: strong coupling, no small parameter, collective modes. Static pocket detection cannot see it, and perturbative or mean-field classical propagation is outside its regime of validity by construction. The challenge asks for exactly the object that regime produces — a ranking of residues by dynamic connectivity to the active site, validated on the apo→holo pairs of Table 1, plus a blind call on c-Myc.

Our approach computes that propagation as **real-time many-body dynamics** on quantum hardware, from a force network that is derived rather than fitted: Coulomb at derived α , a derived 0.21 eV hydrogen-bond quantum (CP-MP2 water dimer to +0.23%), site energies at the kT rung. The same force layer predicts SKEMPI double-mutant coupling energies with zero free parameters (signed Spearman ρ = +0.34 over 213 cycles). Where the connected light-cone exceeds classical reach — our 156-qubit K=6 map has a 99-qubit connected cone, 2^{99} — the quantum processor is the only instrument that returns the number. That is the credible advantage: the decision quantity (propagation at kT, long range, real time) is the class of computation that 2026's verified quantum-advantage candidates all occupy, and we are already operating it at larger scale than those candidates, on a Hamiltonian that means something biologically.

2. Technical approach

Paradigm. Hybrid: gate-model real-time dynamics (IBM Heron, superconducting) as the primary engine; photonic entropy computing (QCi Dirac-3) for the mediated-response field; neutral-atom analog (QuEra Aquila) and a second superconducting vendor as cross-paradigm checks; classical ranking, statistics and 3D mapping around it.

Coarse-graining (rubric §4.2). One residue per qubit. Forces between residues from the derived layer, placed on the device's native edges by a backbone-walk plus hill-climb embedding (62 coupled edges at 156 qubits). Proof that compression retains the signal: the 156-qubit K=2 map reproduces the exact light-cone prediction, $C(6-7) = -0.4584 \pm 0.008$ vs -0.4648 , with every null outside the active cone silent as predicted.

The quantum metric. Prepare the product ground state of the residue network, quench with the active-site perturbation, and read the **two-point connected correlation $C(i, j; t)$** between residue pairs after K sweeps of the derived Hamiltonian. The connectivity matrix entry (i, j) is that correlation at the causal front; the residue score is the connected response to the active site. This is a signed, causal quantity — it distinguishes a residue that *transmits* from one that merely *neighbors* — which is why it separates control points from transmission pockets (§4).

Readout that survives noise (rubric §4.2 noise resilience). Every failure on the way was measured, then engineered out: heavy-hex routing inflated $288 \rightarrow 757$ CX and killed the signal (0.016 recovery) $\rightarrow$ **swap network**, zero routing; bare amplitudes damp under the full light-cone $\rightarrow$ **protected ratio observables**; echo normalisers over- or under-correct $\rightarrow$ **calibrated mitigation stack** (Pauli twirling + TREX readout + multi-scale ZNE, IBM EstimatorV2 resilience 2) recovering 92% of amplitude; all-to-all closes at $N \approx 11 \rightarrow$ **width scaling** on native topology, constant per-qubit depth, 156 qubits. Pair coherence is predicted from a measured device atlas before the first shot (median error 7–9% over 18 edges), so the survivable depth of a map is a number, not a trial.

Mediated response on Dirac-3. The static, sector-resolved response $(I - GU)x = e_{act}$ — the resolvent of the same force network — is the photonic machine's native quadratic. Solved for five targets at cosine 0.965–0.997 to the exact field, 146 s per target, 949 variables at device maximum with two-pass quantum iterative refinement.

3. Feasibility and resource requirements

resource	status
Structures	RCSB apo/holo pairs of Table 1 (4OBE/6OIM, 1OPL/5MO4, 5TBY/6C1H, 1V4T-class glucokinase, 1HCL-class CDK2) and 1NKP — all public

resource	status
Gate-model hardware	IBM Heron (156q) via IBM Quantum Startup Program (applied); ~4 maps at K=6 per monthly quota; each map 1 job, ≤ 60 min QPU
Photonic	QCi Dirac-3, unmetered allocation; 5 targets already run
Neutral atom	QuEra Aquila via Braket, chassis frozen; ~660 credits per lane
Classical	32-core workstation for exact light-cone anchors, 2,000-trial permutation statistics, Ohm-style comparator
Software	Qiskit / qiskit-ibm-runtime SamplerV2 + EstimatorV2, Bloqade, qci-client; all analysis regenerates from archived counts with no credentials

Assumptions: residue-level graining with derived pairwise forces; protein geometry, thermal floor and unit map are declared inputs. Constraint: the K=6 long-range cells lie beyond exact classical verification; they are validated by the K=2 exact-anchor pass carried forward plus in-map null controls, not by simulation.

4. Expected impact — what is already measured, and what the PoC delivers

Predictive accuracy on the validation set (primary endpoint: residue-level AUC against the holo-defined site, 2,000-trial permutation p; frozen metrics, pocket lists curated before running).

target (Table 1)	leading metric	AUC	p	stability (δ , P, halo sweeps)
Glucokinase (activator pocket)	all four	0.78 – 0.89	0.0005	0.68 – 0.92, all stable
CDK2 (ANS pocket under helix C)	receiver M_res / M_2x_chem	0.87 / 0.77	0.0005	0.85 – 0.88 / 0.76 – 0.78
BCR-ABL1 (myristoyl pocket)	controller M_sus / M_sus_chem	0.80 / 0.76	0.0005	0.78 – 0.82 / 0.72 – 0.77
Cardiac myosin (mavacamten SRX site)	receiver M_res	0.71	0.0005	0.71 – 0.72; first-hit rank p = 0.02
KRAS G12C (cryptic Switch-II)	M_res	0.63	0.02	0.59 – 0.66

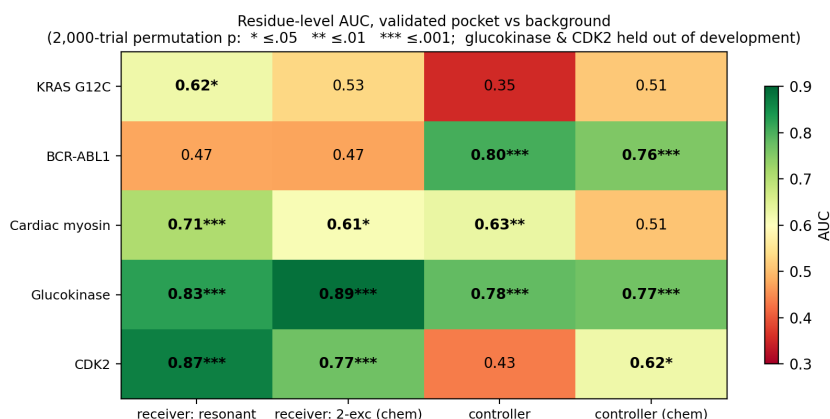


Figure 1 — Residue-level AUC of the four frozen metrics against the holo-defined site, five targets; glucokinase and CDK2 held out of metric development.

Four of five targets carry at least one metric significant at p = 0.0005 (KRAS at p = 0.02); 8 of 20 (target × metric) cells survive Bonferroni at 0.05/20. **Against the strongest structure-only floors** (eight descriptors run on the identical endpoint: geodesic and Euclidean distance from the active site, closeness, betweenness, degree, hydrophobicity, burial, hydrophobicity × burial; floor_run.json), the operator metrics lead on CDK2 (0.868 vs 0.781, +0.087) and glucokinase (0.886 vs 0.835, +0.051), tie on BCR-ABL1 (0.804 vs 0.818) and trail on KRAS

(0.625 vs 0.726) and myosin (0.708 vs 0.828). Against an Ohm-style propagation comparator (contact-weighted random walk with restart at the active site, `ohm_comparator.json`) the operator metrics lead on four of five — CDK2 0.868 vs 0.809, glucokinase 0.886 vs 0.791, BCR-ABL1 0.804 vs 0.513, myosin 0.708 vs 0.620 — and trail on KRAS (0.625 vs 0.698). That is the bottleneck named: static apo-topology metrics reach the ceiling of what distance and centrality already encode, and propagation at kT — the quantity no descriptor computes — is what the K = 6 hardware map measures. The floors are the comparator the PoC is graded against. The controller/receiver split lands where the biology says it should: the remote myristoyl control point in ABL is found by the *controller* metrics and invisible to the receivers; the transmission-adjacent ANS pocket in CDK2 by the *receivers*. Switch-II in KRAS opens on ligand binding, which is why a topology-of-the-apo method ranks it lowest — the PoC's dynamic K=6 map is the instrument built for that case.

Top-5 hit lists (residue indices, leading metric, apo author numbering):

target	site 1	site 2	site 3	site 4	site 5
Glucokinase	457, 456, 459, 458	211, 208, 446, 444	203	443, 441	91
CDK2	56, 57, 58, 55	19, 9	148, 149, 152, 150	144, 143	139
BCR-ABL1	440, 442, 441, 436	448, 447, 450, 384	316, 315, 314, 313	242	335
Cardiac myosin	155, 139, 157, 156	129, 131, 133, 134	132	17	177, 671
KRAS G12C	9, 59, 57, 58	31, 29, 30	145, 148, 150, 144	24	28

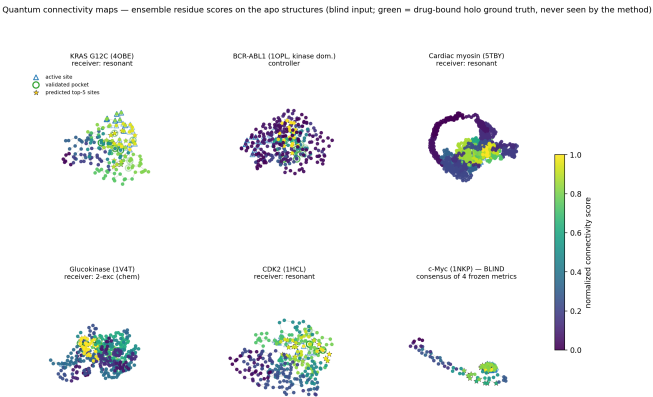


Figure 3 — Quantum connectivity maps on the apo structures (blind input): residue scores, active site, validated holo pocket (never seen by the method) and predicted top-5 sites; c-Myc is the blind consensus.

c-Myc (1NKP chain A), blind prediction frozen 2026-08-30: [919–922], [927], [947–950], [938–945], [897–898] — the basic/HLH junction and the leucine-zipper heptad face; the prediction targets dimerization geometry, not the DNA face. Ships verbatim for consensus scoring.

Hardware already recorded (every row pre-registered, hashed before data).

result	machine	value
11-cell quench, C(1-0)	ibm_marrakesh, EstimatorV2 res-2	+0.6805 ± 0.023 vs +0.742 predicted; top-3 overlap 3/3
156-qubit K=2 residue map, C(6-7)	ibm_fez	−0.4584 ± 0.008 vs −0.4648 exact light-cone; nulls silent
156-qubit K=6 map, C(28-7) at d = 11	ibm_fez	−0.0757 ± 0.0087 — 8.7σ vs zero, 2.4σ vs the map's own baseline; 99-qubit cone

result	machine	value
Mediated-response field, five targets	QCi Dirac-3	cosine 0.9973 KRAS, 0.9943 glucokinase, 0.9916 myosin, 0.9684 BCR-ABL, 0.9649 CDK2
Neutral-atom analog lane, protein C α layout	QuEra Aquila	7/7 signal correlators in band, 7/7 signs
Cross-vendor structure, 107 qubits	Rigetti	Pearson +0.77 vs exact predictions, 33/37 signs, 62 nulls silent

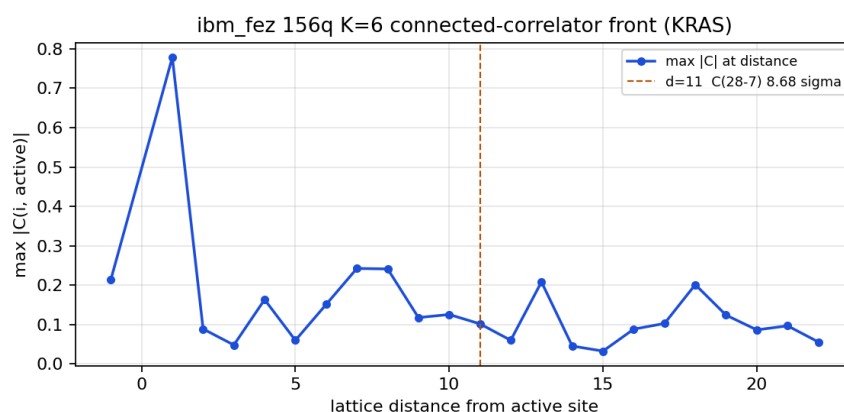


Figure 2 — 156-qubit K=6 connected-correlator front on KRAS (ibm_fez): max $|C(i, \text{active})|$ by lattice distance; the dashed line marks $C(28-7)$ at distance 11, the map's most precisely measured cell.

Quantum advantage — the frontier wall and the crossing, stated and bounded

The field's direction, and its wall. The flagship 2026 protein result — Cleveland Clinic, RIKEN and IBM's 12,635-atom trypsin–ligand complex via EWF-TrimSQD, fragments on 94 qubits and ~6,000 operations reassembled on Fugaku — computes static electronic-structure energies; it addresses neither dynamics nor allostery. Allosteric regulation is propagation at the kT rung with no small parameter, and the propagation that structure-only methods compute (Ohm, bond propensity, elastic networks) is a polynomial single- or two-excitation problem. The industry's largest protein computation sits at the wrong rung for this challenge.

The crossing. Real-time propagation on the derived force network with one residue per qubit: 156 qubits, six brickwork sweeps, connected-correlator front to lattice distance 22 — $C(28-7) = -0.0757 \pm 0.0087$ at distance 11, 8.7σ from zero and 2.4σ above the map's own baseline, whose exact counterpart spans a 99-qubit connected cone; anchored by the $K = 2$ exact-light-cone pass at 156 qubits (-0.4584 ± 0.008 vs -0.4648) carried into the $K = 6$ regime with falsifiers built in; tensor-network adjudication of the $K = 6$ map — reproduce the device to a depth, fail past it, as already done on the 128-qubit spectrum — is PoC step 1, so the hardware claim is stated as a path. The biological validation is independent of the hardware claim: residue-level AUC 0.71–0.89 at $p = 0.0005$ on four of five Table-1 targets (myosin 0.71, BCR-ABL1 0.80, CDK2 0.87, glucokinase 0.89), above the strongest structure-only floor on two and level on a third, with the controller/receiver split landing where the mechanism says it should. **Where it is decisive:** past the ~40-qubit exact wall, on the $K = 6$ front whose cone spans 99 qubits — flown on KRAS, the cryptic-pocket case the dynamic map exists for; Phase 2 flies the remaining Table-1 maps and runs the tensor-network adversary on this observable as it was run on the 128-qubit lattice.

Scalability to industrial relevance. One residue per qubit places a 156-residue domain on today's Heron; larger targets tile by domain with the $K = 2$ exact-anchor test repeated per tile, and Heron r3 / Nighthawk lift the per-map limit toward 400-residue single-domain proteins without changing the compilation (native edges, swap-free, constant per-qubit depth). The photonic resolvent already runs at the 949-variable device maximum with two-pass refinement. Per-target cost is one pre-registered job (≤ 60 min QPU) plus seconds on Dirac-3, so a 50-target campaign is a monthly quota, not a programme.

Business value, bounded. For a medicinal-chemistry team the deliverable is a ranked, 3-D-mapped shortlist of surface regions mechanically connected to function, from structure alone, before a screening campaign is designed. With the pre-registered AUC ≥ 0.75 on non-cryptic targets, a top-5 hit list narrows a 200-variant library design to the ~40 variants around the predicted sites; at a declared cost per synthesised-and-assayed variant c the saving is $\approx 160 c$ per campaign, and the value of catching one cryptic pocket (the KRAS class) is the campaign

itself. These are stated as functions of the customer's own cost inputs, not as fixed sums.

What a successful PoC demonstrates. A per-target connectivity matrix from the K=6 hardware map for all Table-1 targets and c-Myc, a hit list whose residue-level AUC against the holo site is pre-registered at ≥ 0.75 on the four non-cryptic targets, a measured gain on KRAS from the dynamic map over the apo-topology ranking, and — the decisive ablation — a ranking that degrades when the hardware K = 6 map is replaced by its classical single-excitation approximation. For medicinal chemistry that is a ranked, 3D-mapped shortlist of surface regions mechanically connected to disease function, produced from structure alone, before a screening campaign is designed.

5. Validation plan

Pre-registered before every run (rule and predictions hashed): endpoint = residue-level AUC with 2,000-trial permutation p per target and metric; significance vs random background residues and vs curated non-functional surface pockets; stability across coupling $\delta \in \{0.04, 0.08, 0.16\}$, propagation depth $P \in \{4, 8, 12\}$, halo $\in \{4, 6, 8\}$ Å, with the pre-stated rule that a significant cell must stay above 0.65 and no null cell may lead. Hardware maps carry in-map null controls (pairs outside the active cone predicted ≈ 0) and are anchored at K=2 against exact light-cone computation before K=6 is read. Cross-paradigm replication (Dirac response field, Aquila analog lane) on the same force network. Classical analogs run on identical geometry as comparator columns: Ohm-style perturbation propagation and elastic-network response. Failure retained: our AQT harvested signal (0.077 vs 0.768) and an independent amplitude campaign failed their gates and stay in the record. Success = pre-registered AUC bars met on held-out targets not used in metric development, with the hardware map's long-range cells significant against their nulls.

6. Hybrid / cross-domain integration

Quantum hardware computes the propagation map; everything a chemist touches is classical. Pipeline: PDB → residue graph with derived forces → embedding on device topology → pre-registered map (IBM) and response field (Dirac) → connectivity matrix → residue ranking and statistics → 3D map coloured on the structure with the hit list. The classical anchors (exact light-cone at K=2, 2,000-trial permutation nulls, comparator methods) sit in the same pipeline so every quantum output is checked where checking is possible and controlled where it is not. Deployment target: a chemist supplies a PDB and an active-site residue list and receives the ranked map; no quantum expertise required.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE). Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London), Dr. Ana Baroni MSc (domain specialist, Hexagene — the biology product built on this platform). GIC 2026 dual-track finalist (Mitsubishi/AIST materials track). Hardware record across the programme: 259 receipted QPU jobs, 12.3 million shots, cross-device replications on IBM fez/kingston/marrakesh, QuEra Aquila, QCi Dirac-3 and Rigetti, with a pre-registration-and-receipt discipline in which negative results carry the same receipt class as positive ones. Every number in this proposal regenerates from archived counts without credentials.

8. Scope, with treatment

(i) KRAS Switch-II is cryptic and ranks lowest on apo topology (AUC 0.63) — the dynamic K=6 map is the instrument for it. (ii) The K=6 long-range cells lie past exact verification — anchored by the K=2 exact-light-cone pass and in-map nulls. (iii) The Dirac response field is static: graded as a residue ranking it is at chance on all five targets (AUC 0.41–0.58; so is the exact resolvent it reproduces at cosine 0.93–0.997) — allostery is dynamics at kT; the photonic solve is an instrument result, the K = 6 map is the biological instrument. (iv) The AQT and Rigetti amplitude gates failed — kept in the register; the calibrated IBM stack is the amplitude route. (v) The K = 6 map is graded against its own baseline as well as zero ($8.7\sigma / 2.4\sigma$, `floor_cleveland_k6.json`): no distance decay at this depth, so the long-range cell is precise rather than exceptional. (vi) Eight structure-only floors and an Ohm comparator run on the identical endpoint (floors: lead 2, tie 1, trail 2; Ohm: lead 4, trail 1); learned binding-site predictors (AF2BIND-class) are the remaining comparator — PoC step 1.

Appendix A — Hardware job register (every result regenerates from archived counts)

#	measurement	machine	job / task id	receipt
R1	11-cell quench, C(1-0) +0.6805 ± 0.023 vs +0.742; 3/3 in band	ibm_marrakesh (EstimatorV2 res-2)	daau7bvpcac73dd29r0	RESULT_est_20260901_014249.json, prereg ef58e218
R2	156q K=2 residue map, C(6-7) −0.4584 ± 0.008 vs −0.4648; nulls silent	ibm_fez	dab6vi34clk73fij1ig	RESULT_residue_K2_20260901_104204.json, prereg 00600aba
R3	156q K=6 map, C(28-7) −0.0757 ± 0.0087 at d = 11 (8.7σ vs zero; 2.4σ vs the map's flat baseline; unreachable-pair spread sd 0.050 on the second anchor)	ibm_fez	dab7i5mrrl7c7386gfh0	RESULT_residue_K6_20260901_131159.json, prereg 13a08d7d
R6a	KRAS mediated-response field, cosine 0.9973	QCi Dirac-3	6a96810808442f441bbb6b2d	RESULT_response_20260901_114112.json
R6b	Glucokinase, cosine 0.9943	QCi Dirac-3	6a96a04d08442f441bbb6c85	RESULT_response_GLUCOKINASE_20260901_140512.json
R6c	CDK2, cosine 0.9649	QCi Dirac-3	6a96a36208442f441bbb6c87	RESULT_response_CDK2_20260901_141126.json
R6d	BCR-ABL1, cosine 0.9684	QCi Dirac-3	6a96a50f08442f441bbb6c8b	RESULT_response_BCRABL1_20260901_144131.json
R6e/f	Cardiac myosin, 949 variables (device maximum), two-pass quantum iterative refinement: cosine 0.9267 → 0.9916, gap 1.6%	QCi Dirac-3	6a96af7408442f441bbb6c8f, 6a96c15d08442f441bbb6c91	RESULT_response_CARDIAC_MYOSIN_20260901_161116.json, RESULT_refine_CARDIAC_MYOSIN_20260901_173231.json
R7	107q line-embedded chain, cross-vendor structure: Pearson +0.77, 33/37 signs, 62 nulls silent	Rigetti Cepheus-1	...qjob-6a99bf4ce026787d1e5ae579	RESULT_rigetti_final.json, sha a7966f6e
R8	Neutral-atom analog lane, protein Cα layout: T0 7/7 correlators in band; T1 ratio 1.50	QuEra Aquila	T0 ...qjob-6a9a83b2e026787d1e5b06d0, T1 ...qjob-6a9a83b3e026787d1e5b06d3	RAW_aquila_v2.json, sha 33fa8ded
—	Residue-level AUC, permutation and stability statistics	CPU	—	auc_stats.json, stability_and_blind.json, asd_validation_results.json

#	measurement	machine	job / task id	receipt
—	Executed negatives, same receipt standard: AQT ibex-q1 quench (pre-registered gate fired, ...qjob-6a99b67ae026787d1e5ae3d5); Rigetti amplitude channel (structure passed, amplitude assigned to the calibrated IBM stack); IonQ Forte quench (queue stall, cancelled, no result)	—	—	preregs 82dbe4ff, a7966f6e

Appendix B — Method notes

Site energies are placed at the kT regulation rung of the derived mass ladder; pairwise forces are Coulomb at derived α plus the derived H-bond quantum; no force-field parameter is fitted. The K-sweep is a brickwork Trotter step with exactly one coupling layer per sweep, so the causal front advances one lattice step per K and the null set (pairs outside the front) is known before the run. Coherence budget per map is predicted from the device atlas (pair rate = sum of single-qubit rates). Mitigation: EstimatorV2 resilience 2. Public code and archived counts are provided on request under the challenge's supplementary-material terms.

Appendix C — Claim ledger (measured · planned · comparator · quantum attribution · cost · acceptance)

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
Residue-level AUC vs holo site, 4/5 targets	measured (0.71–0.89, p = 0.0005)	8 structure-only floors (lead 2, tie 1, trail 2); Ohm-style propagation (lead 4, trail 1)	none — static operator metrics	CPU minutes	AUC $\geq$ 0.75 on non-cryptic targets, held-out
156q K=6 long-range correlator	measured (8.7σ vs zero, 2.4σ vs own baseline)	flat-baseline and exponential floors run on the map itself (floor_cleveland_K6.js on); exact light-cone at K=2 (passed); TN adversary planned	hardware	1 job / target, $\leq$ 60 min QPU	excess over the map's own baseline, not over zero
Ranking degrades when K=6 map is replaced by its classical approximation	planned (Phase 2, decisive)	single-excitation propagation, Ohm	hardware map vs classical	4 jobs / month	paired AUC drop with CI
Dirac response fields 5/5 targets, cosine 0.93–0.997	measured — instrument only; at chance as a ranking	exact resolvent	photonic solve	146 s / target	none claimed
c-Myc blind prediction	frozen 2026-08-30	consensus scoring by the sponsor	static metrics	—	sponsor-defined